quantmsdiann: a scalable SDRF-driven DIA-NN workflow for reanalysis of single-cell, spatial, and bulk proteomics datasets

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Abstract

Public proteomics archives now host thousands of data-independent acquisition (DIA) datasets, but they remain difficult to integrate and reuse because each dataset has been processed with a different software configuration, and most lack standardised metadata. Here, we present `quantmsdiann`, an open-

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source Nextflow/nf-core workflow that parallelises DIA-NN across cloud and HPC infrastructures, guided by the experimental design declared in SDRF format. The workflow ships container-pinned profiles for each supported DIA-NN version (extensible to any release via a custom container) through Docker and Singularity, with no manual installation or dependency resolution, and natively handles all vendor formats, converting to mzML when needed (e.g. SCIEX). It exports harmonised quantification tables as MSstats input, the new Quantitative Proteomics eXchange format (QPX), and a pmultiqc quality-control report. The parallelisation design reanalyses a 2,300-run single-cell dataset in 2.4 hours on an HPC cluster of 300 nodes. We benchmark quantmsdiann on the full ProteoBench DIA panel (five DIA-NN versions × four instrument modalities) and reanalyse public DIA deposits spanning bulk cell lines (NCI-60/PXD003539, ProCan/PXD030304), a single-cell oocyte plexDIA dataset (MSV000093870), a spatial Deep Visual Proteomics cohort (PXD064049), and a phosphoproteomics cohort (PXD049692). SDRF-driven reprocessing with current DIA-NN versions matches or exceeds the originally deposited bulk and single-cell results, recovers comparable precursor-level depth on the spatial cohort, and matches a directDIA analysis at the phosphopeptide-backbone level; five cell-line cohorts are further combined into a pan-cohort under one invocation. quantmsdiann is a step towards scalable, reproducible reanalysis of the growing DIA data landscape and a foundation for community efforts in large-scale DIA meta-analysis and atlas building. quantmsdiann is available at https://github.com/bigbio/quantmsdiann.

Keywords: DIA-NN, single-cell proteomics, spatial proteomics, phosphoproteomics, plexDIA, data-independent acquisition, Nextflow, nf-core, SDRF, reproducibility, QPX, pmultiqc

1. Introduction

Data-independent acquisition (DIA) is now the dominant mass-spectrometry strategy for proteomics at scale, and DIA-NN [1] is among the most widely adopted analysis engines across timsTOF [2, 3], Astral, and Orbitrap platforms. Recent benchmarks confirm DIA-NN's competitiveness for single-cell DIA workflows [4], and recent releases have expanded support for plexDIA, timsTOF parallel accumulation-serial fragmentation (PASEF), and vendor-native raw-file reading. Despite this analytical maturity, most DIA-NN users run the tool as a desktop application, which limits its scalability for large datasets. Deployments on HPC and cloud infrastructure remain difficult, relying on shell scripts, bespoke directory conventions, and hand-curated sample sheets. As a result, the same raw data analysed by two laboratories, or with two `DIA-NN` releases, can yield non-overlapping quantification tables that are difficult to reconcile.

Beyond per-study analysis, the value of public proteomics archives such as PRIDE Archive [5] and other ProteomeXchange partners [6] lies as much in systematic reanalysis as in original deposition. In 2024 we published bigbio/quantms [7], a Nextflow pipeline that standardised multi-engine DIA and DDA analyses behind a Sample and Data Relationship Format (SDRF) [8] input on the nf-core framework [9]. quantms was, however, built around `DIA-NN` 1.8.1 with a fixed parameter set, and its DDA/DIA breadth made extending DIA-specific functionality difficult. New `DIA-NN` parameters conflicted with DDA options, versions were hard-coded, sample-level acquisition parameters were not properly propagated to per-run `DIA-NN` invocations, and raw-file conversion was mandatory. Iterative reanalysis at archive scale requires a workflow that combines the reproducibility of nf-core, the sensitivity of current `DIA-NN` releases, the metadata discipline of SDRF, and a parallelisation strategy capable of processing thousands of raw files.

Here, we present `quantmsdiann`, an independent SDRF-driven Nextflow pipeline dedicated to `DIA-NN` that extends DIA-specific features beyond what the original bigbio/quantms pipeline could accommodate. While `quantmsdiann` shares its SDRF-first design philosophy and parts of the configuration-generation tooling with quantms, its workflow, container set, and module library are developed and released independently, with container-pinned profiles for every `DIA-NN` version selectable at runtime under Docker or Singularity. Downstream, it exports a harmonised QPX (Quantitative Proteomics eXchange) Parquet archive alongside the standard `DIA-NN` reports and MSstats input (Methods). We demonstrate the pipeline along three axes: ProteoBench [10] benchmarks across four DIA modalities and successive `DIA-NN` releases; independent reanalyses of public DIA deposits spanning bulk cell lines (NCI-60, ProCan), a single-cell oocyte plexDIA dataset, a spatial Deep Visual Proteomics cohort, and a phosphoproteomics cohort; and a pan-cohort of five public DIA cell-line reanalyses integrated under a single SDRF-driven invocation.

2. Experimental Procedures

2.1. Pipeline implementation

quantmsdiann is implemented in Nextflow DSL2 ($\geq$25.10.4) and follows nf-core community standards [9] for pipeline structure, testing, documentation, and continuous integration. It supports both DIA-NN library-free analysis (predicting the spectral library in-silico from a FASTA reference) and library-based analysis against a user-supplied spectral library; all analyses in this study used the library-free mode with the UniProt human reference proteome [11] (UP000005640, downloaded March 2026). Each computational step is packaged as a versioned container image (Docker, Singularity, or Apptainer). Open-source community tools (ThermoRawFileParser [12], pridepy [13], QPX, and the wiffconverter SCIEX reader) are pulled from public registries (BioContainers [14] or ghcr.io/bigbio). Because the DIA-NN license restricts redistribution, only version 1.8.1 is shipped as an image (the default, from BioContainers); releases from 1.9 onward (1.9.2–2.5.1, including the 2.5.1 enterprise build) are built locally from the per-release recipes in the companion quantms-containers repository (https://github.com/bigbio/quantms-containers) and tagged so the workflow selects them automatically (Supplementary Note 2). The repository at https://github.com/bigbio/quantmsdiann is organised as a standard nf-core pipeline with main workflow (main.nf), subworkflows, modules, test data and CI configurations, and documentation. Pipeline version v2.1.0 (release codename "Sao Pablo", 2026-05-08) was used for the analyses reported in this paper.

2.2. SDRF parsing and configuration generation

quantmsdiann accepts SDRF [8] as primary input. SDRF parsing is performed using the sdrf-pipelines Python package developed within the quantms project, which validates the SDRF against the proteomics-specific subset of EFO, PSI-MS, and PRIDE ontology terms and emits a normalised per-sample channel for downstream Nextflow processes. DIA-NN configuration generation extracts enzyme, fixed and variable modification, instrument, precursor mass tolerance, fragment mass tolerance, and missed-cleavage settings from the SDRF and writes a DIA-NN command-line configuration. Where SDRF columns are absent, DIA-NN defaults appropriate for the inferred instrument are applied.

2.3. Raw file handling and `DIA-NN` execution

Vendor formats are first-class inputs in `quantmsdiann`: unlike quantms, which converts every raw file to mzML, `quantmsdiann` reads most formats natively and converts only when required. Thermo `.raw` files are read natively by `DIA-NN` $\geq$ 2.1.0 (which added Linux native-reader support), or otherwise converted to mzML with ThermoRawFileParser [12] (`-mzml_convert`); Bruker `.d` folders are ingested natively, with compressed forms (`.d.tar`/`.zip`/`.gz`) decompressed first; and Sciex `.wiff` / `.wiff2` are converted to mzML with the open-source `wiffconverter` (`https://github.com/bigbio/wiffconverter`). Generic `.dia` and pre-converted `.mzML` inputs pass through unchanged, with optional OpenMS re-indexing (`-reindex_mzml`).

2.4. Quality control and QPX export

Per-run and per-cohort quality control reports are generated by `pmultiqc` [15], a proteomics extension of MultiQC [16] that summarises `DIA-NN`-specific metrics including peptide and protein counts, missed cleavages, charge-state distributions, precursor and fragment mass accuracy, and per-run identification yield. The QPX archive is produced by the `qpxc` CLI from the `qpx` package (`https://github.com/bigbio/qpx`), which writes Parquet files for identifications (psm, peptide, protein-group, feature), metadata (sample, run, ontology), and provenance, and optionally exports MuData (`.h5mu`) containers for scverse-based [17] downstream analysis. The aim of exporting to QPX is to bundle the SDRF, identifications, quantifications, and pipeline provenance into a single queryable artefact for reanalysis and long-term archival, and to provide a standardised input for downstream tools that support the format (e.g. MSstats, Perseus, or custom scripts).

2.5. Datasets benchmarked and compute environment

Supplementary Note 1 (Table S1) lists all benchmark and reanalysis datasets with their instrument, acquisition mode, and deposition year: the ProteoBench DIA modules 5/7/9/10, the three per-cohort cell-line reanalyses (PXD003539, PXD004701, PXD030304), the two additional pan-cohort datasets (PXD041421, PXD017199), the PXD064049 spatial Deep Visual Proteomics cohort, and the PXD071075 scalability sweep; it also provides each dataset's SDRF prepared from the original sample metadata. For each cell-line reanalysis cohort, the original published quantification was downloaded from PRIDE as the comparator, with both it and the `quantmsdiann` reanalysis thresholded at 1% precursor-level FDR; the conservative contaminant/decoy filter and the like-for-like-threshold rule applied to the original headline counts are described in Supplementary Note 3. Runtime benchmarks were run on the EMBL-EBI HPC cluster (LSF scheduler), with per-task CPU and memory dispatched by Nextflow and memory profiled from the Nextflow trace report (`-with-trace`).

3. Results

3.1. `quantmsdiann` *pipeline architecture*

Built on the nf-core framework [9], `quantmsdiann` accepts a single SDRF file as primary input, which encodes per-sample metadata (cell line or tissue origin, factor values, instrument, post-translational modifications) and per-run file references. We extended the SDRF with DIA-relevant columns (MS1 and MS2 mass windows, and plexDIA channel definitions and labels) and added an adapter to sdrf-pipelines [8] (`convert-diann`) that translates each run's declared metadata into a per-run `DIA-NN` command-line configuration. Every run is therefore parametrised from its own metadata with no manual configuration or parameter-harmonisation step, removing a common source of analysis-script divergence between groups: different MS1/MS2 window schemes can be mixed within a cohort, or a plexDIA dataset run with its channel definitions, without any workflow change beyond the SDRF. The workflow comprises four mandatory and three optional modules organised around the `DIA-NN` analysis core (Fig. 1a), alternating between per-file parallel stages (red; raw-file preparation, preliminary calibration, and individual search) and collective stages (green; in-silico library generation, empirical library assembly, final quantification, MSstats conversion, and `pmultiqc` [15] quality-control reporting) operating on the full cohort.

`quantmsdiann` follows `DIA-NN`'s distributed multi-pass flow, alternating parallel per-file work with serial cohort-wide steps (Fig. 1a). Each raw file is first prepared *in parallel* (vendor-format conversion, decompression, or indexing) where needed; conversion to mzML is mandatory for early `DIA-NN` versions and for some vendor formats such as SCIEX `.wiff`. A spectral library is then predicted in-silico from the FASTA once (*serial*; skipped when a library is supplied), and every file is calibrated against it in a *parallel* first pass. The first-pass results are merged into a single empirical library (*serial*); each file is searched again against that library *in parallel*; and a final consolidation pass merges all per-file results into the cohort quantification, applying match-between-runs, normalisation, and protein inference (*serial*). MSstats conversion, `pmultiqc` reporting, and QPX export then run once at the end (*serial*). The two per-file search passes carry essentially all of the compute and scale across cluster nodes, driving the throughput reported below (Section 3.2); the serial steps set a largely fixed wall-clock floor. Outputs include the standard `DIA-NN` report tables: `report.tsv` (or `report.parquet`), `report.pr_matrix.tsv`, and `report.pg_matrix.tsv`, alongside MSstats-formatted input and a `pmultiqc` [15] quality-control report. A harmonised QPX archive bundles identifications, metadata, and provenance into a Parquet-based artefact, queried directly with DuckDB and, for the quantification layers, exported to MuData (`.h5mu`) for scverse-compatible [17] downstream analysis.

3.2. quantmsdiann throughput and scaling

To characterise how `quantmsdiann` scales on an HPC cluster, we simulated the reanalysis of an entire single-cell experiment of ~2,300 raw files (PXD071075) across cluster sizes from 10 to 300 nodes (set through the Nextflow `executor.queueSize`), and measured the end-to-end wall-clock time (Fig. 1b). Because the per-file work is parallelised across nodes, a single SDRF-driven invocation completed the whole cohort within hours: wall-clock fell monotonically from 37.4 h at 10 nodes to 2.2 h at 300 nodes (excluding the one-time in-silico library-prediction step, which is independent of cluster size). The curve has a practical knee near 200 nodes (2.4 h), beyond which the serial cohort-level steps dominate and additional nodes give diminishing returns (a 30× increase in cluster width yields only a 17.4× speed-up), so the cluster should be sized to the desired turn-around time rather than to maximum throughput.

We then compared the time-to-finish across every dataset in this study (Fig. 1c), reporting each run net of the one-time in-silico library-prediction step (a cohort-independent cost that can be predicted once and reused). Across cohorts spanning three orders of magnitude in file count, wall-clock stayed within hours because the per-file passes absorb additional files in parallel: the 5,798-run ProCan cohort (PXD030304) finished in 6.2 h, in the same band as cohorts orders of magnitude smaller, while the 6-run ProteoBench benchmarks completed in under 1.5 h. Time-to-finish is therefore governed by per-run search cost and cohort-level consolidation rather than by the number of files: the longest reanalysis was a 206-run cohort on an older Q Exactive instrument

(PXD017199, 11.4 h), exceeding the far larger ProCan cohort, followed by a 10-file phosphoproteomics cohort whose deep variable-modification search drives its 8.0 h (PXD049692). Peak memory during the cohort-level consolidation stayed within the 64–128 GB available on commodity HPC nodes, so no high-memory specialist hardware is required. Per-step wall-clock and resource envelopes are reported in Supplementary Note 4 (Figs. S1 and S2).

3.3. ProteoBench benchmark across `DIA-NN` versions and instruments

To test whether `quantmsdiann` preserves `DIA-NN`'s analytical performance under SDRF-driven orchestration, we processed every public ProteoBench DIA module through the workflow and compared the resulting identification counts against the ProteoBench public-submission snapshot (29 May 2026) [10]. Because `quantmsdiann` runs `DIA-NN` library-free (predicting the library in-silico from the FASTA, with no externally supplied experimental library), we restricted the community set for a like-for-like comparison to ProteoBench `DIA-NN` submissions that used the same predicted-from-FASTA library (`predictors_library` = DIANN); submissions that loaded a pre-existing experimental or user-curated spectral library search a different candidate space and are excluded from the headline comparison. The benchmark panel covers four instrument families and acquisition modes: Orbitrap Astral 2 Th DIA (*Module 7*), single-cell DIA on Orbitrap Astral (*Module 9*, PXD049412), timsTOF diaPASEF (*Module 5*, PXD062685), and SCIEX ZenoTOF 7600 (*Module 10*, PXD070049). For each module we ran the workflow against three `DIA-NN` configurations — the oldest supported release (1.8.1) and the current release in its academic (2.5.1) and enterprise (2.5.1-enterprise) builds — in a single invocation, all parametrised from the same SDRF (Fig. 2). Identification counts are taken from the per-precursor `DIA-NN` report rather than the quantities matrices. Each release is run at `DIA-NN`'s recommended run-specific precursor q-value (`-qvalue`: 1% for 1.8.1, 5% for the 2.5.1 builds); on top of this we apply, uniformly across configurations, a 1% global precursor q-value, and count unique precursors identified in at least one run. Protein groups are summarised — like-for-like across versions, each at a 1% run-specific protein-group q-value — as the per-run average and the count quantified in *every* run ("complete profiles"); the per-run depth metric tracks a release's gains more directly than the cross-run union, which saturates as rare identifications accumulate across replicates. The recommended run-specific precursor cut-off mainly affects the reproducibility-filtered ($\geq$3-replicate) precursor counts; the single-replicate precursor totals and the per-run protein metrics apply the same thresholds to every version and are directly comparable across releases.

Precursor counts (identified in at least one run) rose from 1.8.1 to 2.5.1 on every module, by 12% on Module 7 (116,771 → 130,246), 11% on Module 9 (41,756 → 46,243), 21% on Module 5 (97,149 → 117,396), and 16% on Module 10 (88,179 → 102,457); the enterprise build adds a further 2–3% on each module. Protein-group depth rose in step. Averaged per run, protein groups at 1% FDR increased from 1.8.1 to 2.5.1 by 8–11% (Module 7 10,091 → 11,167; Module 9 7,157 → 7,895; Module 5 10,199 → 11,231; Module 10 8,788 → 9,470), with the enterprise build a further 1–3% higher and highest on every module. The improvement is larger for proteins quantified in *every* run (complete profiles): the enterprise build gains 12–20% over 1.8.1 (Module 7 8,799 → 10,403, +18%; Module 9 6,124 → 7,349, +20%; complete-profile panel in Supplementary Note 5). Reproducibility-filtered precursor counts (≥3 replicates) follow the same ordering across the three configurations on all four modules. This monotonic version-progress trend is consistent with what the broader ProteoBench community reports across `DIA-NN` releases [10], indicating that the SDRF wrapper introduces no version-coupling artefacts. Quantification accuracy, by contrast, is essentially invariant across `DIA-NN` versions. On every module the measured per-species $\log_2$ fold-changes recover the expected HYE ratios (human at unity, yeast and *E. coli* near their design ratios, with the mild attenuation at the extreme *E. coli* ratio that is characteristic of DIA quantification), and the three `DIA-NN` configurations overlie one another at every species (Fig. 2b). The version-to-version shift in median quantification error $|\epsilon|$ ($\sim$0.01 $\log_2$, $\approx$0.5–1% in linear fold-change) is far smaller than the per-precursor spread, so the choice of `DIA-NN` release is best driven by the identification gains in panel (a) rather than by accuracy. On the two modules for which the snapshot contained predicted-library `DIA-NN` submissions, Orbitrap Astral (Module 7, $n = 15$) and timsTOF diaPASEF (Module 5, $n = 8$), `quantmsdiann`'s three configurations fall within the community median-$|\epsilon|$ distribution, at or below its median (Fig. 2c); the single-cell and ZenoTOF modules carried no predicted-library community submission. Per-version, per-module protein-group and precursor breakdowns are provided in Supplementary Note 5 (Fig. S3).

3.4. Single-cell DIA reanalysis improves with current `DIA-NN` releases

To assess `quantmsdiann` on low-input data, where library-free sensitivity and match-between-runs matter most, we reanalysed two label-free single-cell HeLa cohorts — Orbitrap Astral single-cell (PXD046357) and One-Tip (PXD044991) — through the workflow under `DIA-NN` 1.8.1 and the current 2.5.1-enterprise build, both parametrised from the same SDRF (Fig. 3). The newer build increased total identifications on both cohorts: on HeLa Astral, union precursors rose from 19,674 to 24,111 (+23%) and protein groups from 3,903 to 4,574 (+17%); on One-Tip, precursors rose from 11,618 to 16,534 (+42%) and protein groups from 1,597 to 2,306 (+44%; Fig. 3a), with the gain holding per cell (Fig. 3b). The improvement is largest in the data-completeness profile — the number of protein groups quantified across an increasing fraction of cells (Fig. 3c) — consistent with stronger cross-run propagation, while quantitative precision and dynamic range are preserved (Fig. 3d,e). As an orthogonal, multiplexed test we reanalysed the first single-cell plexDIA cohort processed through `quantmsdiann` (MSV000093870, Galatidou *et al.* 2024 [18]; mTRAQ 3-channel, Q Exactive), quantifying 2,904 protein groups from the same raw data versus 2,122 in the original analysis (+37%). It recovered 94% of the originally reported groups (1,999 of 2,122) and a further 905 groups at comparable per-cell median depth (1,736 versus 1,784, across 107 and 97 cells), with a QC-matched per-oocyte comparison confirming equivalent single-cell depth (Pearson $r = 0.98$; Fig. 3f and Supplementary Note 6).

3.5. Pan-cohort reanalysis of public cell-line DIA deposits

To show the value of running one SDRF-driven workflow across many independent deposits, we reanalysed five public cell-line DIA panels in a single `quantmsdiann` invocation: NCI-60 (PXD003539, Guo *et al.* 2019 [19]; originally OpenSWATH), ProCan-DepMapSanger (PXD030304, Gonçalves *et al.* 2022 [20]; originally `DIA-NN` 1.7.x), the Sun *et al.* 2023 PCT-SWATH breast-cancer panel (PXD004701, 76 lines [21]), the MultiPro batch-effect testbed (PXD041421, Wang *et al.* 2023 [22]), and Tognetti *et al.* 2021 (PXD017199, 67 breast-cancer lines [23]) (Fig. 4). For the two cohorts with a published DIA protein-group headline, the reanalysis recovered more protein groups under the original studies' $\geq$ 2-unique-peptide criterion, applied identically to both sides: NCI-60 6,328 versus 4,284 (+48%) and ProCan 8,166 versus 6,698 (+22%; Fig. 4a). The Sun panel, already analysed with a modern DIA engine, also gained at 1% global FDR (7,412 versus 6,091 protein groups); PXD041421 and PXD017199 carry no published DIA headline and contribute reanalysis-only counts. Across the five cohorts the reanalysis spans 12,845 unique UniProt accessions, of which 5,495 (42.8%) are detected in all five (a pan-cohort core proteome) and a further 1,526 (11.9%) in exactly four. Tissue-level coverage spans 28 pan-cancer tissue categories (Fig. 4b), which maps both the cell lines contributing to each tissue and the union of unique proteins detected per tissue, led by lung, breast, and haematopoietic-and-lymphoid tissue. Because every count comes from a single SDRF-parameterised `quantmsdiann` invocation, the resulting protein matrix is queryable from the published QPX archive and can be filtered, joined, or re-grouped by Cellosaurus, NCIt, or Disease Ontology terms without manual harmonisation. Per-cohort completeness, peptide-per-protein distributions, gene-versus-accession overlap and per-cell-line missingness are provided in Supplementary Note 6 (Figs. S4–S12), which also include an NK-cell Fe-NTA phosphoproteomics reanalysis (PXD049692; comparable phosphopeptide-backbone coverage at 1% FDR). A spatial Deep Visual Proteomics cohort of MYCN-amplified neuroblastoma (PXD064049) was likewise reanalysed and matched the original `DIA-NN` 1.8.1 analysis at the precursor level (16,443 versus 16,518).

4. Discussion

`quantmsdiann` automates the path from a deposited DIA dataset to a reproducible quantification table, at a scale that supports systematic reanalysis of public DIA archives. The SDRF-first design encourages experiment annotation upstream of analysis and produces metadata that is interoperable with the wider proteomics ecosystem [8]; the QPX archive bundles the SDRF, harmonised quantification tables and pipeline provenance into a single Parquet container so that reanalysed results can be archived (e.g. on Zenodo) and re-queried alongside the original raw data. By focusing on `DIA-NN` rather than the multi-engine scope of the related bigbio/quantms pipeline [7], `quantmsdiann` achieves a leaner dependency surface and faster pipeline-level execution for the DIA-only regime, where the dominant cost is per-file `DIA-NN` invocation rather than across-engine comparison.

This parallel structure also shapes how the workflow scales. End-to-end wall-clock is the sum of a largely fixed serial overhead (chiefly in-silico spectral-library generation from the FASTA, the single largest non-parallel step, together with cohort-level consolidation) and the per-file analysis stages, which run as independent tasks across the cluster (Fig. 1b) and whose elapsed time grows only sub-linearly with cohort size as files are processed in parallel waves (per-step envelopes in Supplementary Note 4, Fig. S1). Time-to-finish is therefore dominated by the fixed overhead for small cohorts and stays within a few hours even for the largest deposits, so a small dataset is not

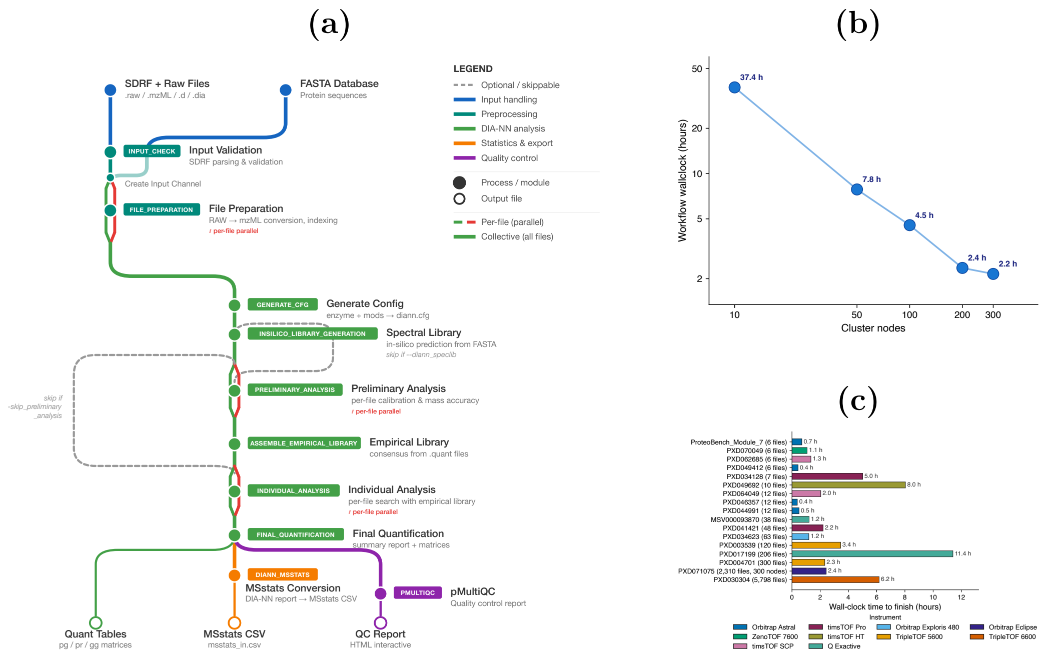

Figure 1: **quantmsdiann pipeline architecture and runtime scaling.** (**a**) Subway diagram of the quantmsdiann workflow: mandatory modules as filled circles, optional/skippable modules as open circles. Per-file parallel stages (red; file preparation, preliminary calibration, individual search) run as one task per raw file, whereas collective stages (green; in-silico library generation, empirical library assembly, final quantification, MSstats conversion, and pmultiqc reporting) run once over the full cohort. (**b**) Workflow wall-clock time versus the number of cluster nodes (Nextflow executor.queueSize) on the PXD071075 single-cell cohort. (**c**) Wall-clock time to finish each reanalysis, one bar per dataset, ordered by cohort size (number of MS data files) and coloured by instrument family. In both (**b**) and (**c**) the reported time *excludes* the in-silico spectral-library prediction step (INSILICO_LIBRARY_GENERATION), a one-time, cohort-independent cost that can be predicted once and reused, so the bars reflect the per-file quantification work. Time-to-finish stays within a few hours across two orders of magnitude in file count because per-file stages are parallelised across the cluster; the deeper variable-modification search of the phosphoproteomics cohorts (e.g. PXD049692) is the main residual driver of wall-clock once library prediction is set aside.

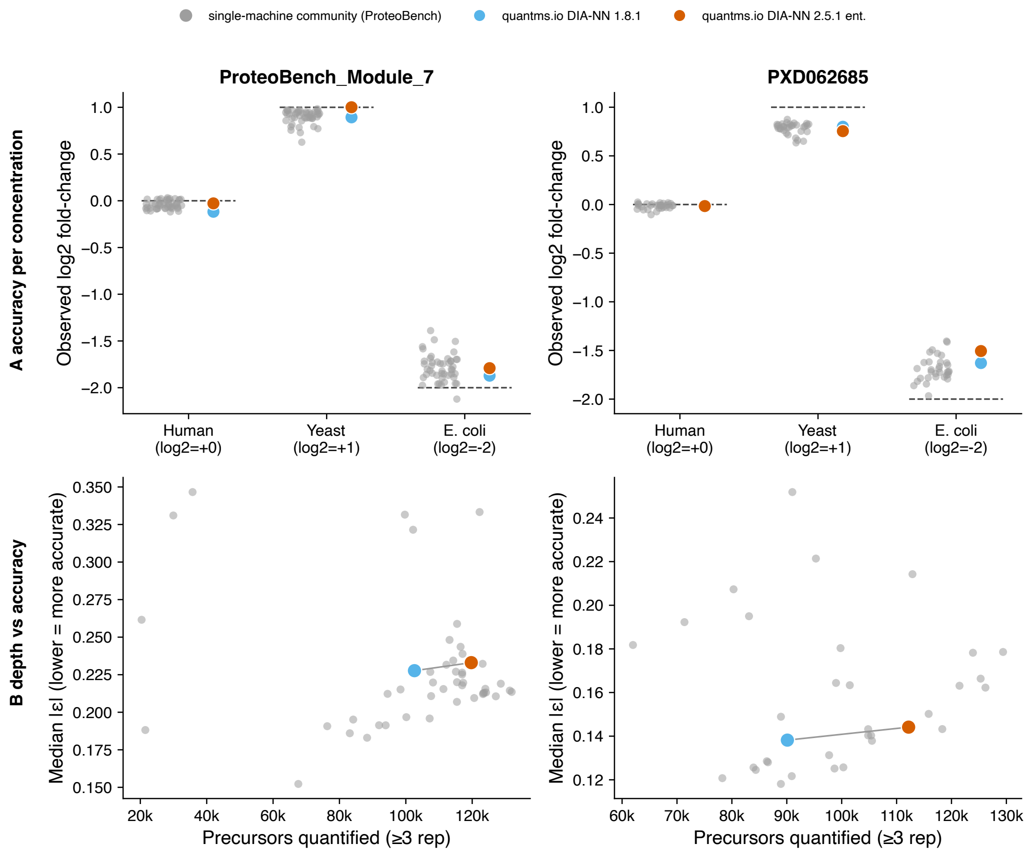

Figure 2: **Distributed `quantmsdiann` reproduces single-machine results on ProteoBench.** For the two DIA modules with public predicted-from-FASTA community submissions (Orbitrap Astral, Module 7; timsTOF diaPASEF, PXD062685), `quantmsdiann` (`DIA-NN` 1.8.1 and 2.5.1-enterprise) is compared against the cloud of community submissions, which are single-machine sequential runs. `quantmsdiann` preserves match-between-runs by design: a preliminary pass builds a combined empirical library across all runs, against which each run is re-quantified. (**a**) Accuracy per concentration — measured $\log_2$ fold-change for each HYE species against the ProteoBench-expected ratio (dashed); community submissions as grey points. (**b**) Depth versus accuracy — precursors quantified against median-$|\epsilon|$; `quantmsdiann` (coloured) sits within the single-machine community cloud (grey), reaching greater depth at comparable accuracy from 1.8.1 to 2.5.1-enterprise. Datasets without predicted-library community comparators (single-cell Module 9/PXD049412; ZenoTOF Module 10/PXD070049) and the per-version protein-group counts are in Supplementary Note 5.

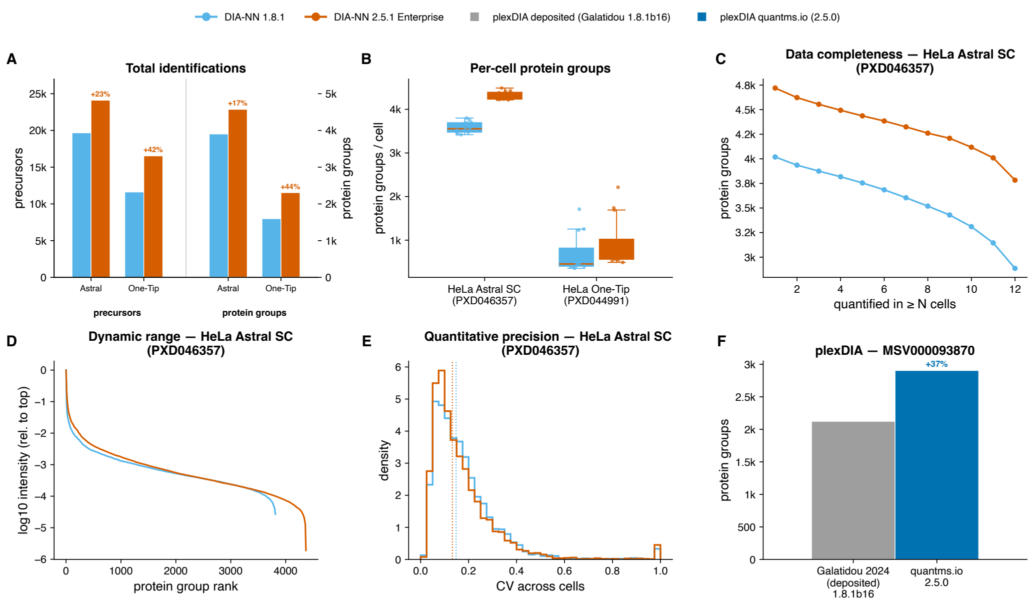

Figure 3: **Single-cell DIA reanalysis by `quantmsdiann`, comparing `DIA-NN` 1.8.1 with the current 2.5.1-enterprise build on identical raw data.** Two label-free single-cell HeLa cohorts are shown — Orbitrap Astral single-cell (PXD046357) and One-Tip (PXD044991) — alongside an orthogonal multiplexed comparison. (**a**) Total identifications: union precursors and protein groups across all cells, per build. (**b**) Protein groups per cell (box and jittered points), showing the newer build lifts depth in every cell. (**c**) Data-completeness curve (HeLa Astral): protein groups quantified in at least N cells — the match-between-runs signature, where the right endpoint is the complete-profile count. (**d**) Relative rank-abundance (HeLa Astral), each run normalised to its most-abundant protein, showing extension into lower abundance. (**e**) Distribution of per-protein coefficient of variation across cells (HeLa Astral), summarising quantitative precision. (**f**) Multiplexed single-cell plexDIA (MSV000093870, Galatidou 2024; mTRAQ 3-plex): protein groups for the deposited analysis versus the `quantmsdiann` reanalysis of the same raw data.

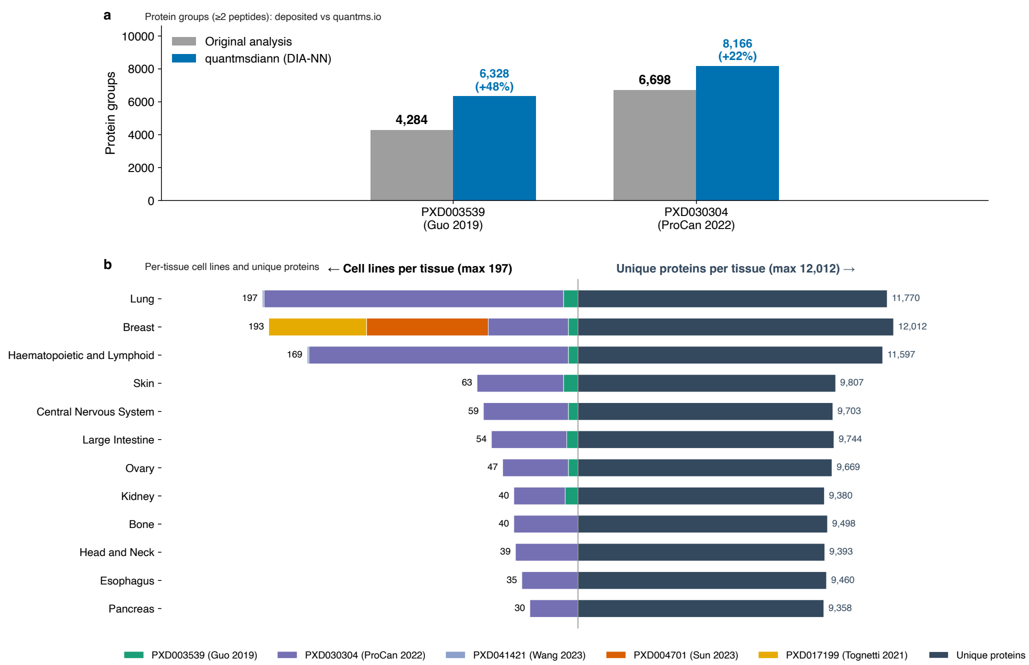

Figure 4: **Pan-cohort of five public DIA cell-line reanalyses by `quantmsdiann`.** Combines PXD003539 (NCI-60, Guo 2019), PXD030304 (ProCan-DepMapSanger, 949 lines), PXD004701 (Sun 2023, 76 breast cancer lines), PXD041421 (Wang 2023), and PXD017199 (Tognetti 2021, 67 breast cancer lines), reanalysed under a single `quantmsdiann` invocation. (**a**) For the two cohorts with a published DIA protein-group headline (NCI-60 and ProCan), protein groups requiring ≥ 2 unique peptides: the originally published count (grey) and the `quantmsdiann` reanalysis (blue), with the same filter applied to both sides. (**b**) Back-to-back per-tissue summary across the combined panel: cell lines contributing to each tissue, stacked by cohort (left), and the union of unique UniProt proteins detected per tissue (right).

necessarily faster than a much larger one (Fig. 1c); the serial overhead itself scales with the predicted-library size, growing for wide variable-modification searches such as phosphoproteomics, rather than with the number of files.

The demonstrations stress the workflow along three complementary aspects: analytical fidelity (ProteoBench; Fig. 2), recovery against published per-study quantifications (independent reanalyses; Figs. 3, 4), and archive-scale throughput (pan-cohort; Fig. 4). The recovery results suggest that the value of DIA reanalysis is real but time-limited (largest for deposits originally analysed with older or non-`DIA-NN` engines and shrinking toward parity for recently processed cohorts), a regime an SDRF-driven workflow is suited to address as new `DIA-NN` versions ship. The workflow already supports broader single-cell and laser-microdissected spatial acquisition modes (Fig. 1a, library-handling branch), and SDRFs for the Brunner [2], Derks [24], Leduc [25], Wang [4], Mund [26] and Makhmut [27] datasets are in preparation.

The cross-`DIA-NN`-version reproducibility view enabled by `quantmsdiann` is, to our knowledge, the first pipeline-level treatment of this question for `DIA-NN`. We expect this to become increasingly important as `DIA-NN` evolves: a deposited single-cell DIA dataset analysed with `DIA-NN` 1.8 today and re-analysed with `DIA-NN` 2.x in two years should not produce different biological conclusions for the same data and parameters. In current practice such re-analyses are rarely performed, because the bespoke per-study scripts make them prohibitively expensive, a barrier the single-invocation audit of Section 3.3 removes.

`quantmsdiann` has several limitations. First, `quantmsdiann` does not currently include batch correction, cell-type assignment, or differential-abundance modules; these are best handled in downstream R or Python packages such as scp [28], which can read the QPX or MSstats outputs directly. Second, library generation for plexDIA channel deconvolution remains internal to `DIA-NN`, and `quantmsdiann` inherits any limitations of that step. Third, the QPX format will continue to evolve as community feedback accumulates; backward compatibility is maintained through schema versioning. Fourth, the single-cell, spatial, and phosphoproteomics demonstrations here (the oocyte plexDIA, the MYCN-amplified neuroblastoma Deep Visual Proteomics, and the NK-cell phospho cohorts) are proof-of-principle; broader benchmarking across these modalities (Fig. 1a, library-handling branch) remains future work.

Finally, `quantmsdiann` complements existing single-cell DIA workflows rather than competing with them. The benchmarking work of Wang *et al.* [4] compared multiple search engines and rescoring strategies; `quantmsdiann` adopts the broadly competitive DIA-NN engine and focuses on the orthogonal problem of orchestrating that engine reproducibly at scale. Similarly, earlier nf-core DIA efforts such as DIAproteomics [29] and, more recently, the MHCquant2 pipeline [30] address analogous orchestration problems for DIA and immunopeptidomics. Together, these contributions extend the nf-core proteomics ecosystem to cover the major MS-based proteomics modalities behind a common SDRF-driven interface.

Data Availability

All datasets reanalysed in this study are publicly available through ProteoMeXchange. Cell-line bulk DIA reanalyses: PRIDE PXD003539 (Guo 2019, NCI-60), PXD004701 (Sun 2023), PXD030304 (ProCan-DepMapSanger), PXD017199 (Tognetti 2021), PXD041421 (Wang 2023). Spatial Deep Visual Proteomics reanalysis: PRIDE PXD064049 (MYCN-amplified neuroblastoma, diaPASEF). Scalability experiment: PRIDE PXD071075 (Wu 2025). ProteoBench benchmarks: PXD049412 (single-cell, Module 9), PXD062685 (diaPASEF, Module 5), PXD070049 (ZenoTOF, Module 10), and Module 7 (Astral) datasets. The full list of PRIDE accessions used in the pan-cohort of DIA cell-line reanalyses is provided in Supplementary Note 1 (Table S1). `quantmsdiann` is available at `https://github.com/bigbio/quantmsdiann` under the MIT license; documentation is hosted at `https://quantmsdiann.quantms.org/`. The pipeline is implemented in Nextflow DSL2 ($\geq$25.10.4) and packages each step as a versioned Docker, Singularity, or Apptainer container image. `DIA-NN` 1.8.1 is distributed publicly via BioContainers; because the `DIA-NN` license restricts redistribution of later releases, containers for `DIA-NN` 1.9 and above are built locally from the recipes in `https://github.com/bigbio/quantms-containers` (see Experimental Procedures). Pipeline version v2.1.0 (release codename "Sao Pablo", 2026-05-08) was used for the analyses reported in this paper. Analysis scripts that generated the figures in this manuscript are available at `https://github.com/bigbio/quantmsdiann-paper` (this repository).

Supplemental Data

This article contains supplemental data. The Supplementary Notes provide the full PRIDE accession list (Supplementary Note 1, Table S1), `DIA-NN` container-build recipes (Note 2), the filtering and protein-counting rules

(Note 3), per-process runtime and resource envelopes (Note 4, Figs. S1–S2), the ProteoBench per-version identification breakdown (Note 5, Fig. S3), and per-cohort completeness, peptide-per-protein and gene-versus-accession overlap figures (Note 6, Figs. S4–S12).

Conflict of Interest

The authors declare no competing interests.

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Author Contributions

QXY, YS and CD developed the pipeline. HW helped build the containers and contributed to the workflow implementation and manuscript writing. AL annotated the SDRF metadata and helped run the workflow. TS, FC, NS and JN assisted with the DIA-NN reanalyses and manuscript writing; TS additionally contributed to the workflow design, and JN tested the tool and identified multiple issues. OK provided feedback on the plexDIA datasets. VD provided DIA-NN expertise and guidance on cross-version reproducibility. MB contributed to manuscript writing and supervised the students; MB and YPR conceived and supervised the project. QXY and YPR wrote the original draft, with input from all authors. All authors read and approved the final manuscript.

Abbreviations

CLI command-line interface

DDA data-dependent acquisition
DIA data-independent acquisition
DIA-NN data-independent acquisition neural networks (DIA search and quantification software)
diaPASEF data-independent acquisition with parallel accumulation–serial fragmentation
FDR false discovery rate
HPC high-performance computing
HYE human, yeast, and *E. coli* (ProteoBench three-proteome mixture)
IQR interquartile range
LSF Load Sharing Facility (HPC scheduler)
MS mass spectrometry
MS1/MS2 first-/second-stage mass spectrometry (precursor/fragment scans)
PASEF parallel accumulation–serial fragmentation
QPX Quantitative Proteomics eXchange
SDRF Sample and Data Relationship Format

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